William Chuang

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SUMMARY OF QUALIFICATIONS

- **Strategic Thinker (INTJ):** Focused on long-term, high-impact solutions in AI-driven cryptanalysis, OSINT, and cybersecurity.
- **Advanced Math & Physics:** Deep expertise in hyperbolic geometry, Kleinian groups, Galois theory—crucial for secure algorithm design.
- **Robust R&D:** 10+ years at respected institutions (Arizona, SFSU, Penn State, NTU), applying theoretical methods to practical challenges.
- **Security & Cryptography:** Applied curvature, limit sets, and Galois groups in post-quantum encryption and secure data protocols.
- **OSINT & Cybersecurity Research:** Self-studying Open-Source Intelligence (OSINT), pentesting, network forensics, and adversarial ML applications in threat detection.
- **Machine Learning & Data Security:** Applying ML models to system log analysis, TCP/IP traffic inspection, firmware integrity verification, and network anomaly detection.
- **Computational Proficiency:** Python (NumPy, scikit-learn, PyTorch), R, C/C++, Java, Lisp, Mathematica, Bash scripting; reverse-engineering low-level firmware; built large-scale cryptography tools.

EDUCATION

University of Arizona (Expected Spring 2025)
M.S. in Mathematics, advanced Ph.D.-level coursework

San Francisco State University (Spring 2022)
M.A. in Mathematics; Thesis on Schottky groups (Advisor: Dr. C.-K. Lai)

University of San Francisco (Fall 2018)
B.S. in Mathematics, Minor in Computer Science; GPA: 3.88/4.00; *Honors*

CORE COMPETENCIES

Mathematical Cryptography & Security:
Hyperbolic geometry, Kleinian groups, post-quantum cryptography, topological vulnerabilities.

Developing a novel encryption framework leveraging group actions on neural networks rather than solutions of differential equations. This approach, independently conceived and later found to parallel the motivation behind Poincaré's Fuchsian groups and monodromy, restructures security landscapes in the post-quantum era by embedding cryptographic transformations within geometric symmetries.

Data Analysis & Machine Learning:
Transformer architectures (multi-head attention), autoencoders, geometric/topological ML approaches.

Algorithm & HPC Development:
Python/C++ for large-scale data, GPU-accelerated ML, system-level optimization.

Research & Technical Writing:
Multiple publications/presentations; formal proof tools (Lean 4, LLMs).

AI-Augmented Intelligence Analysis & OSINT Automation:
Developing transformer-based (local LLM) AI agents trained on intelligence cycle methodologies (e.g., Heuer's structured analysis, HUMINT techniques) for real-time data fusion, deception detection, and adversarial influence modeling.

Human Factor Security & OSINT:
Defending against psychological manipulation, decoding body language, behavioral profiling, lie detection via baseline analysis, NLP for communication security, countering mind control tactics, and influence operations in intelligence settings.

Cognitive & Influence Security:
Adversarial intelligence analysis, AI-enhanced OSINT gathering, NLP-driven deception detection, psychological profiling via transformer models, and social engineering countermeasures.

RELEVANT RESEARCH EXPERIENCE

University of Arizona (2022–Present)

- Prof. S. Sethuraman: Real analysis; self-studied stochastic processes for cryptanalysis.
- Prof. S. Cherkis: Explored Nahm equations, geometric field theories, and used Lean 4 + LLMs for secure AI.
- Prof. N. Hao: RTG Project on transformer attention scaling.
- Prof. C. Haessig: Investigated corresponding polynomials of Galois groups via Python code; self-studied cryptographic classification.
- Prof. D. Glickenstein: Mentored a project reconstructing Mirzakhani's study on hyperbolic geometry and closed geodesics; self-study on encryption applications in transformer architectures and autoencoders.

San Francisco State University (2019–2022)

- Computed Hausdorff dimension of Schottky groups; applied fractal geometry for data obfuscation.
- Applied the prime geodesic theorem to secure high-dimensional data.

Pennsylvania State University (2017–2018)

- Investigated Hardy's proof of uniform distribution for pseudo-random generation.
- Studied topological invariants with applications in encryption algorithms.

National Taiwan University — LeCosPA (Pre-Baccalaureate, 2011–2013)

- Researched TQFT, AdS/CFT, and vacuum energy; early work in quantum information.

ORIGINAL CONTRIBUTIONS AND HISTORICAL FIRSTS

The following research contributions represent original discoveries, each constituting a historical first in its respective domain, based on available literature and records:

1. Redefining Core Algebra in Neural Networks Using Hyperbolic Geometry:
Unlocks model orbit mobility under group actions—achieving a level of transformation that Minsky (with Papert, 1988) showed intractable in classical perceptrons.

- **Historical First:** First geometrically consistent, hyperbolically grounded deep learning framework enabling group-based model mobility.
- **Recent Extension:** Framework supports an infinite family of consistent geometric methods preserving expressiveness of Euclidean networks. Allows negative-valued outputs and parameter flows, unified under invariant symmetry operations.

2. First Quantum-Ready Transformer Architecture Built from Geometric Principles:
Each layer corresponds to Möbius transformations from Lie algebra symmetries, evolving via Schrödinger-type dynamics.

3. Monodromy of Neural Networks:
Extends classical monodromy to global evolution of deep learning models as parameters traverse loops—capturing topological changes in model behavior.

4. Topological Entropy as a Cryptographic Hardness Parameter:
Introduced Hausdorff dimension δ as a tunable constant governing cryptographic complexity and obfuscation under classical and quantum threats.

5. Neural Encryption via the Holographic Principle:
Mapped transformer components to holographic dual structures:

- Input tokens $\leftrightarrow$ boundary observables
- Transformer layers $\leftrightarrow$ bulk fields
- Weight dynamics $\leftrightarrow$ RG flow
- Cipher inversion $\leftrightarrow$ bulk geometry reconstruction
- Semantic representation $\leftrightarrow$ automorphic kernel functions

Encryption requires full latent structure, making inversion equivalent to solving holographic reconstruction problems.

6. AdS/CFT-Inspired Neural Cryptosystem:
Implemented bulk geometry in $\mathbb{H}^{n+1}$ with transformer layers as bulk fields and outputs as boundary observables. Security modeled analogously to an "S++ class" based on duality inversion hardness.

- Duality is symbolic, not metaphysical—implemented via tensor structure and geodesic kernels.
- Architecture simulates renormalization flow and bulk-boundary encoding with attention based on partition functions.

7. Embedding Langlands Duality in Neural Cryptographic Design:
Developed GPT-style model embedding Langlands duality into training dynamics. No known prior AI or crypto models formalize this.

8. First Use of Automorphic Forms in Attention Mechanisms:
Augmented attention with automorphic functions, aligning language representations with harmonic analysis.

9. Loss Landscape Control via Lorentz Transformations:
Used Lorentz transformations to systematically reshape energy landscapes as a geometric alternative to gradient descent.

10. Modular Spectral Indistinguishability Assumption (MSIA):
Proposed a post-quantum cryptographic primitive based on symbolic dynamics and zeta function inversion:

$$\zeta_p(z) = \frac{1}{\det(I - zA(p))}, \quad A(p) \in \mathbb{F}_p^{n\times n}$$

- **Security Model:** Theoretically analogous to NP-hard and #P-hard problems.
- **Classification:** S+ to S++-class depending on symbolic obfuscation layers.
- **Historical First:** First to use modular trace zeta functions and symbolic modules in cryptographic hardness assumptions.

11. First Explicit Cascade Model for Laboratory Vacuum Energy Amplification:
Constructed a fully consistent theoretical framework for creating controlled vacuum energy gradients in condensed matter systems by combining Coulomb and nuclear slingshot amplification, gravitational Schwinger mechanisms, and Lee–Yang criticality diagnostics.

12. Traceless Vacuum Burst Stress–Energy Tensor as Backreaction Regulator:
Proved that engineered vacuum bursts yield a traceless stress–energy tensor to leading order, ensuring that gravitational backreaction remains perturbative. Provided analytic verification under Einstein field equations with semiclassical corrections.

13. First Integration of Lee–Yang Criticality with Quantum Neural Amplification Cascades:
Demonstrated that quantum amplification cascades sharpen vacuum instabilities at critical field thresholds, embedding criticality within operator dynamics. Extended Lee–Yang zero formalism to engineer tunable, localized vacuum phase transitions.

14. Quantum Path Integral Construction for Vacuum Energy Harvesting:
Linked Euclidean quantum-gravitational path integrals with experimental vacuum engineering protocols. Modeled controlled barrier tunneling and amplification cascades as critical instanton processes embedded within engineered condensed matter arrays.

15. Laboratory-Compatible Curvature-Driven Schwinger Amplification Model:
Developed the first explicit model showing how condensed matter structures, through coherent mechanical or acoustic driving, can transiently approach gravitational Schwinger thresholds without requiring Planck-scale energies.

INDEPENDENT MATHEMATICAL THEORIES IN NEURAL ARCHITECTURE, CRYPTOGRAPHY, AND MATHEMATICAL PHYSICS

The following original contributions satisfy formal requirements for a mathematical theory: explicitly defined objects, internal axioms, morphisms or dynamics, and closure under composition. These frameworks generalize, unify, or predict structural phenomena in deep learning, post-quantum cryptography, and quantum field theory. Many admit categorical formulations and support modular extension.

1. GRAIL (Geometric Representation Algebra for Intelligent Learning) — Universal Meta-Architecture for Geometry-Based Neural Computation

- GRAIL encodes neural computation as algebraic operations over curved manifolds (e.g., hyperbolic, Lorentzian, modular).
- Representation-theoretic linear functionals-based implementations represent a finite subclass within GRAIL. GRAIL supports an infinite variety of implementations, encompassing geometric, symbolic, entropic, and cryptographic constructions. Inner product–based methods—though compatible with curved-space learning—constitute only a narrow, algebraically constrained subclass within this broader operational landscape.
- GRAIL more broadly admits nonlinear, non-symmetric, non-metric operations derived from automorphic kernels, quantum deformations, and symbolic-entropic dynamics.
- These generalized operators support encryption, symbolic attention, post-quantum obfuscation, and arithmetic alignment, subsuming and extending all traditional deep learning frameworks.

- Defines neural computation over general curved manifolds (hyperbolic, Lorentzian, elliptic).
- Generalizes to future neural geometry architectures beyond Euclidean space.
- **Training Regimes:**
  - (1) Backprop-compatible curved-space operators, suitable for gradient-based optimization.
  - (2) Symbolic, non-differentiable operators (e.g., Langlands layers, cryptographic kernels) trained via monodromy, Lorentzian landscape reshaping, fixed-point iteration, or categorical dynamics.
- **Satisfies:** Generalized geometric axioms, symmetry group closure, nonlinear composition, categorical morphism consistency.

2. First-Principle Quantum Neural Dynamics (FPQND) — Theory of Quantum Neural Evolution

- Establishes Schrödinger-type evolution equations from Lie symmetries and Hamiltonians.
- **Satisfies:** Axiomatized operator evolution, Lie algebra generators, tensor flow conservation, category-consistent time dynamics.

3. Langlands–Neural Correspondence (LNC) — Categorical Theory of Arithmetic–Neural Duality

- Maps automorphic and Galois representations to training dynamics.
- Defines functorial updates replacing gradient descent using arithmetic spectral data.
- **Satisfies:** Langlands functoriality, trace axioms, Frobenius morphisms, dual group correspondence.
- **Derived Architecture:** Quantum–Langlands Attention Integration (QLAI) using Maass forms and Hecke spectral filters.

4. Monodromic Training Dynamics (MTD) — Theory of Topological Model Shifts

- Uses monodromy and homotopy group actions to encode training via parameter loops.
- **Satisfies:** Loop consistency, covering space deformation, commutative path dynamics.

5. Curved-Space Neural Encryption (CNE) — Cryptographic Duality over Bulk–Boundary Architectures

- General framework using bulk–boundary dualities for encryption.
- Extends AdS/CFT-inspired systems to all GRAIL-manifolds.
- **Satisfies:** Geometric injectivity, dual-layer hardness, functorial latent encodings.
- Hyperbolic neural encryption is a special case; formulation is geometry-agnostic.

6. Modular Spectral Indistinguishability Assumption (MSIA) — Post-Quantum Cryptographic Theory

- Uses symbolic dynamics, modular traces, zeta matrices, and spectral representations.
- **Satisfies:** Modular matrix algebra, entropy-preserving trapdoors, symbolic representation theory.

7. Fractal Cryptographic Dynamics (FCD) — Entropic Obfuscation via Symbolic Chaos

- Embeds Hausdorff dimension and symbolic entropy into cryptographic constructions.

- **Satisfies:** Measure-theoretic invariance, chaotic symbolic encoding, dimensional tunability.

8. Arithmetic–Neural Quantum Field Theory (ANQFT) — All-Loop Integral Representation via Neural Objects

- Encodes quantum periods and Feynman integrals using neural-algebraic automorphic forms.
- **Satisfies:** Functorial neural mappings, decomposition of modular periods, categorical simplification via neural moduli.

9. Cosmic Neural Galois Representation (CNGR) — Arithmetic Symmetries in Learnable Systems

- Defines a category of neural-motivic objects encoding mixed motives, Galois actions, and period structures.
- Constructs Frobenius-compatible updates, symbolic descent mechanisms, and spectral L-function embeddings.
- **Satisfies:** Functorial Galois realizations, motivic descent rules, categorical representation of arithmetic periods.
- **Applications:** Symbolic arithmetic learning, automorphic AI alignment, period inference, and post-quantum resilience.

10. Vacuum–Graviton Cascade Formalism (VGCF) — Axiomatic Framework for Controlled Vacuum Instabilities

- Formalized vacuum bursts as field excitations governed by curvature-induced criticality and engineered amplification cascades.
- **Satisfies:** Stress–energy tracelessness, semiclassical backreaction suppression, critical curvature tuning, and amplification coherence.
- Includes rigorous derivations linking Coulomb slingshot, nuclear slingshot, and gravitational Schwinger thresholds.

11. Dynamic Lee–Yang Vacuum Engineering (DLYVE) — Quantum–Critical Construction of Vacuum Gradients

- Embeds Lee–Yang zeros and critical partition structures within engineered condensed matter setups.
- **Satisfies:** Analytic continuation of partition functions, critical field tuning, semiclassical–quantum transition diagnostics.
- Enables laboratory scaling of quantum vacuum instabilities without requiring cosmic-scale curvature.

12. Quantum Amplification Cascade Theory (QACT) — Operator-Level Framework for Vacuum Burst Coherence

- Extends classical cascade models to fully quantum regimes using operator algebra and entangled amplification channels.
- **Satisfies:** Quantum-corrected stress–energy conservation, generalized coherence conditions, vacuum phase transition predictability.

NATIONAL SECURITY ALIGNMENT AND CLEARANCE READINESS

The theoretical frameworks developed herein—spanning neural geometry, post-quantum cryptography, Langlands duality, monodromy-based training, and hyperbolic transformers—have direct relevance to national security, cryptanalytic resilience, and post-quantum computing strategies.

A portion of this work, titled *"Hyperbolic Autoencoder and Transformer Framework"*, was submitted as part of a formal RFI response to DARPA-SN-25-60 and is provisionally protected under U.S. Patent Application #63/773,441 (filed March 18, 2025). Key protected innovations include:

- Monodromy-driven hyperbolic transformer layers
- Lorentzian energy landscape modulation
- Langlands correspondence embedded within attention mechanisms for post-quantum security

Export Controls: All research complies with U.S. academic and legal frameworks, including ITAR and EAR public disclosure protocols.
Clearance Readiness: Open to future clearance-based collaboration with U.S. agencies (DoD, DARPA, NSA, CIA, FBI). All current work remains unclassified but is applicable to mission-critical domains.

ADDITIONAL RESEARCH PROJECTS

Self-Study: First-Principles Quantum Transformer/Autoencoder Architectures from Geometric Symmetry. Developed a hyperbolic transformer/autoencoder grounded in the geometry of the Poincaré disk and upper half-plane, with weights as isometries from $\mathfrak{sl}(2,\mathbb{R})$. Designed from exact differential equations (e.g., $\frac{dz}{dt} = \kappa(z - z_-)(z - z_+)$) that govern Möbius dynamics, making the architecture naturally quantum-compatible.

- Mapped classical flows to Schrödinger evolution via infinitesimal generators as Hermitian operators.
- Layers correspond to unitary maps induced by $\mathrm{PSL}(2,\mathbb{R})$ symmetry actions.
- Demonstrated that prior quantum transformers lack symmetry principles or geometric grounding.
- Architecture allows quantization via representation theory (e.g., principal series), interpreting pointwise evolution as wavefunction propagation.
- Enables integration of geometry and quantum logic for LLMs and defense platforms.

Potential Applications:

- Foundation for quantum LLMs in intelligence, scientific simulation, and crypto.
- Hyperbolic flows as structured replacements for matrix multiplications.
- Deployment on quantum hardware using Lie groups, not heuristic circuits.
- Quantum benchmark architecture with exact Hamiltonian dynamics.

Self-Study: Semisimple Rings and Radicals in Coding Theory and Cryptography. Based on lectures by Prof. Klaus M. Lux (Spring 2025). Investigated the role of semisimple rings and the Jacobson radical in coding theory and lattice-based cryptography.

- Topics include linear codes over $\mathbb{Z}_4$, group algebras, ring-based cryptosystems (NTRU, Ring-LWE).
- Developed 10 examples showing how nontrivial radicals influence cryptographic vulnerability and algebraic coding properties.

Self-Study: AI-Driven Network & Firmware Security Analysis. Applied ML techniques to system logs, TCP/IP traffic, and firmware integrity. Focused on adversarial ML for detecting firmware- and chip-level anomalies.

- Studied vulnerabilities in chipsets, embedded firmware, AMT systems.
- Used transformer-based models for behavioral anomaly detection.
- Reverse-engineered firmware using disassembly and differential analysis.
- Integrated OSI-layer forensic tools with AI anomaly detection.

Use Case: Vendor-Specific Commands (VSCs) in ESP32 ROM

- Analyzed hidden commands enabling unauthorized access to 1B+ IoT devices.
- Demonstrated potential for persistent infection, remote code execution, and impersonation attacks in BLE/WiFi networks.
- Studied passive traffic signatures, injected payload detection, and firmware-level backdoor mitigation.

Self-Study: RF Signal Analysis for SIGINT & Cybersecurity. Used SDR tools and spectrum analysis to detect unauthorized RF emissions and electromagnetic intrusion attempts.

- Detected IMSI catchers, rogue base stations, backscatter surveillance.
- Studied modulation and propagation signatures of covert RF transmissions.

Self-Study: AI-Driven Cognitive Security & OSINT Defense. Developed transformer-based agents for real-time deception detection, psychological profiling, and adversarial social engineering defense.

- Based on Heuer's structured analysis (ACH) and adversarial influence modeling.
- NLP-driven behavior baselining and falsehood signal tracking.

Self-Study: Narrative Simulation & Strategic Influence Modeling. Built generative frameworks for simulating adversarial narratives and psychological operations using transformers.

- Wargaming narrative shifts, cognitive dominance strategies, and influence operations.
- Developed AI-based red-teaming pipelines for policy vulnerability analysis.

Self-Study: Emerging Sensing Technologies & Counter-Surveillance. Studied non-line-of-sight (NLoS) sensing technologies and their countermeasures.

- Laser and acoustic imaging through walls and occlusions.
- Hyperbolic transformer detection of anomalous signal propagation.
- Applications in dual-use security, privacy, and gesture-based HCI.

Self-Study: Multimodal Environmental Sensing — RF, Ultrasound, IoT Fusion. Combined RF DensePose, ultrasound sensing, and IoT device fusion for privacy-preserving surveillance countermeasures and health/HCI applications.

- **Privacy:** Hyperbolic signal shaping and adversarial signal injection to block unauthorized imaging.
- **Healthcare:** Non-contact monitoring using smartphone ultrasound echoes and WiFi signals.
- **Rescue:** Hybrid sensing for presence detection through debris or smoke.
- **Global AI Surgery:** Built real-time anatomical datasets using passive sensing for AI medical training.
- **HCI + Polygraph:** Multimodal inputs (RF/ultrasound/biometric) for cognitive and deception analysis.
- **Neurotech:** Studied low-intensity focused ultrasound (LIFU) for modulating stress, cognition, and suggestibility.

Self-Study: Transformer-Based Countermeasures for Document Radiation Espionage. Researched EM scanning attacks using terahertz, zeta waves, backscatter, and reflectometry to extract information from physical documents.

- Used transformers to reverse-engineer document exposure from signal artifacts.
- Developed anomaly detection using curvature-sensitive embeddings (hyperbolic transformers).
- Employed SDR tools to detect and visualize scanning transmissions in real time.
- Designed metamaterial shielding and EMI distortion layers guided by AI.
- Formulated inverse problems to infer adversary scanner configuration and content access.

Applications:

- Secure physical document environments (SCIFs, boardrooms, intelligence offices).
- Detection/prevention of radiative intelligence exfiltration.
- Foundation for Prodeo's SIGINT counter-surveillance toolkit.

PRODEO: AI-DRIVEN INTELLIGENCE SOLUTIONS (LAUNCHING JULY 2025)

Founder & Independent Researcher

- Official launch: July 2025. Focused on OSINT, SIGINT, cybersecurity, narrative simulation, cognitive security.
- Developing transformer-based AI for real-time threat intelligence, risk modeling, and influence operations.
- Engineering autonomous AI teams for red-teaming, policy simulation, and multi-domain influence modeling.
- Designing SIGINT platforms with SDR integration for covert transmission detection and RF environment forensics.
- Building anomaly detection frameworks across OSI layers: logs, network, firmware, and RF traffic.

Innovation Highlights:

- Hyperbolic transformers for document shielding and cognitive baseline profiling.
- LLM-powered deception analysis for adversarial interrogation simulation.
- Fusion of cognitive warfare, PsyOps, and narrative AI for national defense applications.
- Non-invasive polygraph enhancement via RF/ultrasound bio-monitoring systems.

Expanded Capabilities:

- Counter-surveillance with laser/NLoS detection, hyperbolic-transformer anomaly analysis, and AI-shielding architectures.
- Plasma/EM-based propulsion defense systems for subsurface and aerospace threats.
- Next-generation projectile research, including neutrino/gravitational-wave sensing for subterranean ISR.
- Radiation-based espionage detection and document shielding design for secure facility construction.
- AI-enabled structured analytic techniques (SATs) and intelligence cycle automation (Heuer methods).
- Proprietary LLM agents trained on HUMINT/SIGINT/OSINT intelligence cycle tasks for red-teaming and decision support.

TEACHING & LEADERSHIP

University of Arizona (2022–Present)
Graduate Teaching Assistant (GTA) for College Algebra and Calculus I/II.
San Francisco State University (2019–2022)
GTA for Calculus courses, emphasizing proof-based learning and exploration.

AWARDS & CERTIFICATIONS

- Protecting God's Children Online Awareness 4.0 — Order of Malta (March 2025)
- MSRI Summer School Nominee (Oxford, Metric Geometry & Geometric Analysis, 2021)
- Information Security Awareness Training — University of Arizona (2023)
- Safety Preparedness Certification — University of Arizona (2023)
- MASS Program Scholarship — Pennsylvania State University (Full Tuition, 2017)
- ACM SIGMOD Service Award (2016)
- Tackling Big Data — MIT CSAIL Workshop Participant (2015)

TECHNICAL SKILLS

Programming & Tools: Python, C/C++, Java, Lisp, R, Mathematica, Shell scripting, Lean 4, Git, LaTeX
Cybersecurity & OSINT:

- AI-assisted network forensics, transformer-based anomaly detection in TCP/IP traffic
- Firmware intrusion detection, reverse engineering, and adversarial ML

System & Network Security:

- Wireshark/TCPDump, Intel AMT reverse engineering, memory vulnerability analysis

Mathematics & Computation:

- Real/complex analysis, topology, measure theory, stochastic processes
- Encryption algorithms, symbolic dynamics, HPC cryptographic toolchains

Radio Frequency (RF) & SIGINT:

- SDR signal analysis, spectrum monitoring, FCC amateur radio prep

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VOLUNTEER AND COMMUNITY ENGAGEMENT

- **Tucson Math Circle** (Oct 2023 – Present): Maintains website and outreach support.
- **Poverello House** (Feb 2025 – Present): Volunteer aid for homeless and underserved.
- **Knights of Columbus Blood Drive** (Mar 2025, upcoming): Service coordination at St. Thomas the Apostle Parish.
- **Math Circle Teaching Assistant — University of San Francisco** (Spring 2017): Mentored young students in mathematical thinking.
- **Hospital Volunteer — Taipei, Taiwan** (2000–2006): Assisted elderly and children with disabilities.
- **Altar Server — Fu Jen Catholic University** (1998–2003): Supported campus liturgies and religious activities.

ADDITIONAL INFORMATION

- **Faith:** Catholic (Confirmed 28 years); Knight of Columbus (CUF exemplification completed); certified in Protecting God's Children (Order of Malta, March 2025).
- **Languages:**
  - **Fluent:** English, Mandarin/Taiwanese
  - **Currently Learning:** Latin, French, Spanish, Italian, Hebrew
- **Memberships:** Pi Mu Epsilon (University of San Francisco)

References available upon request.

Note: Additional documentation, patent application references, and project whitepapers available upon inquiry.